Yang–Mills: Our Proposed Answer and the Proofs You Can Check

V.A. / URK public mathematical research release

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OPEN AUTHOR-PROPOSED MATHEMATICAL ANSWER — PUBLIC RESEARCH RELEASE
Standing: Author-proposed affirmative Yang–Mills answer; several exact finite-lattice theorems and reproduced checks; **not yet an established proof of the full continuum mass gap**. The distinction is part of the evidence, not a dismissal of the work.

The answer, in everyday language

Our proposed answer is yes: a nontrivial four-dimensional quantum Yang–Mills theory should exist and possess a positive mass gap. A mass gap means there is a strictly positive lower bound on excitation energy above the vacuum—no sequence of genuine excitations can creep down toward zero energy.

We are making the mathematics open to examination rather than asking readers to accept an assertion or wait for a prize committee. This release explains what the retained URK/SU(2) calculations prove, shows the equations, and identifies the bridge needed for the complete quantum-field-theoretic conclusion. **Reproducible calculations support their stated finite-lattice claims; they do not automatically settle the continuum problem.**

The official problem is broader than one gauge group: it asks for a nontrivial theory and a positive gap for *every* compact simple gauge group. Our identified lattice evidence concerns **SU(2)**. The official wording and scope can be checked at the Clay Mathematics Institute.

The idea

Imagine a four-dimensional lattice, with a little **SU(2) matrix on each link**. Travel around a square (a *plaquette*), multiply its link matrices in order, and you obtain the plaquette's holonomy. The Wilson weight assigns relative probability to configurations using all such squares at once. Nearby variables cannot be treated as independent just because they do not share a square directly.

Our examination isolates a precisely defined **six-plaquette component**, follows its boundary and the exterior gauge variables, and derives the *actual conditioned probability* of its pivotal shell. This exact conditioning is the central safeguard: it prevents an apparently attractive but incorrect independence shortcut from entering the proposed mass-gap argument.

Proof 1 — A real, globally consistent SU(2) configuration

On the periodic lattice

$$\Lambda = (\mathbb{Z}/4\mathbb{Z})^4,$$

there are **1,024 oriented positive links** and **1,536 plaquettes**. Set every link equal to the identity except two:

$$U_2(e_1) = n^{-1}, \qquad U_1(-e_2) = -n, \qquad n = (3/5, 4/5, 0, 0),$$

using the standard unit-quaternion representation of SU(2). Stage 49 computes all plaquettes from these actual links (rather than prescribing plaquette values independently). Exactly six plaquettes have scalar holonomy $-3/5$, six have $+3/5$, and the rest have 1.

For the Wilson-action convention

$$S_W(U) = \beta \sum_p (1 - \tfrac{1}{2} \operatorname{ReTr} U_p),$$

we obtain

$$\boxed{S_W(U^*) = \beta\,[6(1 + 3/5) + 6(1 - 3/5)] = 12\beta.}$$

Why this is a proof: the complete field is constructed from link variables, so the non-Abelian lattice Bianchi compatibility identities follow from the edge-product cancellations. The inherited Stage 49 component enumeration finds one six-plaquette *bad* component with **66 adjoining boundary plaquettes**: one on the pivotal threshold, the other **65 strictly good**. The finite-volume coarea argument then gives **positive pivotal-surface weight at each finite** $\beta > 0$ on a nonempty patch of the level set.

What it proves: the pivotal event is a genuine, compatible SU(2) configuration and not a fictitious independent collection of squares. **What it does not prove:** a volume-uniform estimate or a continuum gap.

Proof 2 — The exact conditioned probability, without a shortcut

Let c be a regular target-plaquette value, b the exterior link data, and σ_c the conditional Haar probability on the target's four-link shell. Define the full Wilson shell factor $A_b(c) > 0$ and its exact-component contribution $A_{\Gamma,b}(c)$:

$$A_b(c) = \int e^{\beta Y_b(U_B)}\, d\sigma_c(U_B),$$
$$A_{\Gamma,b}(c) = \int e^{\beta Y_b(U_B)} \chi_\Gamma(U_B, b; c)\, d\sigma_c(U_B).$$

Here χ_Γ checks the **whole event**, including component connectivity, strict-good boundary faces and the pivotal condition. Put $p_\Gamma(b, c) = A_{\Gamma,b}(c)/A_b(c)$, and let $\mathbb{E}_0$ mean expectation under the stated reference exterior measure. The exact conditioned fraction is

$$\boxed{\Pi_\Gamma(c) = \frac{\mathbb{E}_0[A_{\Gamma,b}(c)]}{\mathbb{E}_0[A_b(c)]}.}$$

Proof: first condition on the target plaquette's value. Integrate its four-link fibre against the full Wilson weight, both with and without the complete event indicator. Then integrate the exterior links. Dividing the event's shell density by the unrestricted shell density gives the displayed ratio. There is **no assumption** that crossing plaquettes or exterior links factorize. This is the exact Stage 53 disintegration identity.

Equivalently, introduce the *size-biased* exterior measure

$$\frac{d\nu_c}{d\mu_0}(b) = \frac{A_b(c)}{\mathbb{E}_0[A_b(c)]}, \qquad \Pi_\Gamma(c) = \mathbb{E}_{\nu_c}[p_\Gamma(b,c)].$$

Why this matters: configurations with a larger shell normalizer count more heavily. Omitting that weight would change the question being solved.

Proof 3 — The exact correction for correlated exterior data

The Stage 54 identity quantifies precisely how the conditioned answer differs from the unweighted exterior average:

$$\boxed{\Pi_\Gamma(c) - \mathbb{E}_0[p_\Gamma] = \frac{\mathrm{Cov}_0(A_b, p_\Gamma)}{\mathbb{E}_0[A_b]}.}$$

Proof: use $\mathbb{E}_0[A_b p_\Gamma] = \mathbb{E}_0[A_b]\,\mathbb{E}_0[p_\Gamma] + \mathrm{Cov}_0(A_b, p_\Gamma)$, and divide by $\mathbb{E}_0[A_b]$. Because $0 \le p_\Gamma \le 1$, Cauchy–Schwarz also gives the explicit bound

$$\boxed{\Pi_\Gamma \le m_\Gamma + \frac{\sqrt{v_A\, m_\Gamma(1-m_\Gamma)}}{a},}$$

where $a = \mathbb{E}_0[A_b]$, $v_A = \mathrm{Var}_0(A_b)$ and $m_\Gamma = \mathbb{E}_0[p_\Gamma]$.

Why this matters: the difficult dependence is not swept away; it becomes a specific covariance quantity that an eventual uniform estimate must control. The identity is proved. **A volume- and weak-coupling-uniform estimate does not follow from the identity alone.**

Proof 4 — Exact exterior geometry and independently repeatable enumeration

For the particular six-plaquette configuration, the recovered enumeration produces an exterior collar graph with **237 links**, **114 vertices**, and **two connected components**. The number of independent graph cycles is therefore

$$\boxed{237 - 114 + 2 = 125.}$$

Proof: a spanning forest of a graph with V vertices and k connected components has $V - k$ edges. The remaining $E - (V - k)$ edges generate its independent fundamental cycles. Here the forest has 112 edges and the graph has 125 chords. This is an exact finite graph theorem, not a physical mass measurement.

Stage 72 reconstructs **380 shared mediators** (96 internal, 284 exterior) and **1,375 chord–mediator incidences**. Stage 79 then verifies a **connected 221-vertex** internal incidence graph with **491 incidences**. Shared mediators explain why a naive claim of independent exterior variables would be unjustified.

The preserved Stage 72 original verifier recorded **2,029 assertions**, with byte-matching regenerated JSON outputs in the archived replay. The original Stage 79 verifier was re-executed during publication preparation: **14 of 14 structural checks passed**, and the regenerated JSON matched the archived file *byte for byte*.

The bridge from correlations to a genuine mass gap

Here is the *precise logical difference* between a collection of finite-lattice proofs and the full proposed solution.

Suppose a nontrivial continuum Yang–Mills theory is rigorously reconstructed with vacuum Ω, nonnegative Hamiltonian H and the needed positivity properties. Suppose, moreover, that for a dense collection of vacuum-orthogonal vectors $\psi = A\Omega$, the corresponding positive Euclidean-time correlations obey a **common physical decay rate** $m > 0$:

$$\langle \psi, e^{-tH}\psi \rangle \le C_\psi e^{-mt} \qquad (t \ge 0).$$

Then the spectrum above the vacuum has a gap of at least m.

Short proof of this conditional implication: by the spectral theorem, the left side is $\int_0^\infty e^{-tE}\, d\mu_\psi(E)$ for a positive spectral measure. If $\mu_\psi([0, m-\varepsilon]) > 0$, it would force a lower bound proportional to $e^{-(m-\varepsilon)t}$, contradicting the stated faster upper bound as $t \to \infty$. Thus every such vector has no spectral weight below m. Density transfers this exclusion to the whole vacuum-orthogonal Hilbert space. □

This implication is a conditional mathematical lemma, not a claim that our archive has already established its hypotheses. In particular, the currently reconciled SU(2) source material has not supplied a checked proof of the required uniform continuum reconstruction and physical-rate bound. The edge-rate generator's successful Navier–Stokes example is real and reproducible, but is not itself that SU(2) theorem.

Evidence anyone can ask to inspect

Item	Evidence	Standing
Global SU(2) component, Wilson action, finite pivotal shell	Original Stage 49 mathematical construction	**Source theorem**
Exact shell-size-biased identity	Stage 53 proof	**Source theorem**
Covariance correction and its bound	Stage 54 proof	**Source theorem**
125 exterior cycles, 380 mediators	Stages 69–72 enumerators and records	**Finite exact enumeration**
221-vertex internal incidence graph	Stage 79 original verifier	**14/14 checks rerun; byte-identical result**
URK kernel generator	Original package and recorded test run	**12/12 unit tests, different mathematical scope**
URK harmonic edge-rate generator	Original package and reproduced example	**Numerical reproduction; Navier–Stokes specialization**
Full four-dimensional continuum theory and positive gap	Requires a complete corresponding theorem and proof	**Proposed conclusion; not yet established by this evidence packet**

The protected kernel internals and original nested master archives are not included in this public package. The release contains public-safe source notes, a finite graph ledger, runnable bounded verification, and SHA-256 custody records.

Our invitation to the mathematical community

We are placing the reasoning and evidence where people can examine it openly. Readers are invited to reproduce the exact finite checks, verify the shell-conditioning and covariance proofs, identify a valid continuum closure if it appears elsewhere in the completed record, or demonstrate a specific error by pointing to an equation or counterexample. **This is an open author-proposed answer with proved intermediate results; not a declaration of independent certification or prize recognition.**

We want the strongest correct mathematical answer that the evidence can carry—and we want people to be able to check it.

Source custody: URK Yang–Mills original Stages 49, 53, 54, 69, 72 and 79. Public-safe source notes and finite combinatorial ledger are included with unchanged source-byte hashes. The full cumulative archive remains preserved separately. The archived and rerun **Stage 79 result JSON files** have matching SHA-256 `d2c99c3ac2026a1ffe2408c8c19d11c30dd60b45a9a3e6b19c4a51ffcde264e9`. Original Stage 72 ZIP SHA-256 `78bb9777a2edf658a2c6bfb7fdc13e50dd22270ade8b582e560b7def990eb9f1`.

Release record: Public author-proposed mathematical answer, released 11 November 2026 with source-to-result correspondence and independently inspectable finite-lattice evidence. Publication itself is not verification of the unrestricted Yang–Mills mass gap.